

JIALI ZHANG

EDUCATION

China Agricultural University, Beijing, China

June 2024

- *MS in Basic Veterinary Medicine*
- *GPA: 3.58/4.0*
- *Relevant Coursework:* Advances of Animal Pathology (92/100), Advanced Cell Biology (88.4/100), Vertebrate Embryology and Reproductive Engineering (90/100), Immunohistochemistry (90/100), Functional Morphology (93/100), Techniques of Electron Microscopy in Life science (92/100)

Jilin Agricultural University, Chang Chun, Jilin, China

June 2021

- *BS in Veterinary Medicine*
- *GPA: 4.13/5.0 (Rank 1/171)*
- *Relevant Coursework:* Organic Chemistry (97/100), Animal Pathological Diagnostics (94/100), Animal Pathological Anatomy (94/100), Medical Statistics (94/100), Animal Biochemistry (94/100)

TECHNICAL SKILLS

- *Laboratory Skills:* WB, IF staining, qPCR, IHC staining, ELISA, 2D cell culture, Organoid culture, Single cell isolation, Exosome extraction and characterization, Cryostat sectioning, Confocal microscopy, Transmission Electron Microscope (TEM), mouse disease model establishment, oral gavage, intraperitoneal injection.
- *Programming and data analysis:* 10x Genomic Xenium, GraphPad Prism, SPSS, ImageJ/Fiji, QuPath, SnapGene.
- *Certificates:* Licensed Veterinarian (People's Republic of China) (2022)

RESEARCH EXPERIENCE

Yale University, *postgraduate associate*, New Haven, CT

October 2024- Present

PI: Won Jae Huh, Assistant Professor

Mechanism of Lrig3 Knockout in Chief Cells on Spasmolytic Polypeptide-Expressing Metaplasia Formation (SPEM)

- Generated a tamoxifen-inducible, chief cell-specific *Lrig3* knockout mouse model and demonstrated using WB and IF staining that *Lrig3* loss induces spasmolytic polypeptide-expressing metaplasia (SPEM).
- Induced chief cell-specific expression of *EGFR-L858R* using tamoxifen and doxycycline and assessed the effect of EGFR activation on SPEM development in *Lrig3*-deficient mice using WB and IF staining.
- Inhibited EGFR signaling in chief cell-specific *Lrig3* knockout mice using anti-EGFR antibodies (P1X/P2X) and demonstrated that EGFR inhibition rescued the SPEM phenotype, as assessed by WB and IF staining.
- Verified signaling downstream of EGFR in the chief cell-specific *Lrig3* knockout model and demonstrated using WB and IHC staining that STAT3 activation mediates SPEM development.
- Treated gastric organoids with STAT3 inhibitor and demonstrated that STAT3 inhibition reversed the SPEM phenotype.
- Presented results at Digestive Disease Week (DDW) 2026.

Investigating the Crosstalk Between Notch and Hedgehog Signaling in Ménétrier's Disease (MD)

- Established a mouse model of Ménétrier's disease (MD) by inducing chief cell-specific expression of constitutively active *EGFR-L858R* with tamoxifen and doxycycline and characterized the resulting MD-like phenotype using WB and IF staining.
- Treated MD patient-derived and normal human gastric organoids with the Notch inhibitor DBZ and the Hedgehog inhibitor sonidegib, demonstrating that inhibition of either pathway reversed MD-associated phenotypes, as assessed by IF staining.
- Demonstrated increased NICD expression in tissue samples from patients with Ménétrier's disease and in an MD mouse model using IF staining.

- Analyzed and interpreted experimental data and drafted the manuscript
- Manuscript in preparation.

Developed a Rapid, Nonconventional Approach to Establishing a High-Fat Mouse Model.

- Developed time-course mouse models of metabolic dysfunction using doxycycline alone or in combination with a high-fat diet.
- Evaluated and compared lipid accumulation across multiple time points using Oil Red O staining
- Isolated hepatocytes and analyzed single-cell RNA sequencing data from samples collected at multiple time points.
- Manuscript in preparation.

China Agricultural University, graduate student, Beijing, China

September 2021-June 2024

PI: Yulan Dong, Professor.

Mechanism of Peony Seed Oil (PSO) as Therapeutic Agent in NAFLD (2023-2024)

- Led the design and established a NAFLD mouse model and administered graded doses of PSO via daily oral gavage, demonstrating that PSO alleviates NAFLD in a dose-dependent manner.
- Investigated the main anti-inflammatory mechanism of PSO through WB, IHC and IF staining and demonstrated that PSO attenuates inflammation via the TLR4/NF- κ B pathway in NAFLD.
- Identified α -linolenic acid as the principal component of PSO via GC-MS profiling, suggesting its role in enhancing hepatic antioxidant capacity.
- Validated the effects of α -linolenic acid in HepG2 cells through oil red O staining, IF staining and WB, demonstrating reduced lipid accumulation, decreased inflammation, and enhanced antioxidant capacity via activation of the Keap1/Nrf2/HO-1 pathway.
- Presented results on the 22nd Academic Symposium, 2023.

Ginger derived-Extracellular vesicles Encapsulated Pioglitazone Delivery System for NAFLD (2021-2022)

- Led the design and isolated extracellular vesicles and characterized their morphology, demonstrating a spherical shape with an average diameter of 200 nm.
- Published results as first author on Cell Commun Signal, 2024.

Jilin Agricultural University, undergraduate student, Chang Chun, Jilin, China

September 2017-June 2021

PI: Yongfeng, Professor

Pathogen Detection in Commercial Goose Populations

- Identified pathogen from clinical samples to determine disease type in commercial goose farms.
- Proposed optimal antibiotic treatment strategies for clients based on disc diffusion susceptibility testing.

PUBLICATIONS AND PRESENTATIONS

• **Publications:**

- Shin M, Gabriel TT, Ofosu FK, **Zhang J**, Wang TT, Bechard ME, Coffey RJ, Huh WJ. Transcriptomic profiling reveals the role of Hedgehog signaling as a biomarker and in the pathogenesis of Ménétrier's disease. *J Pathol.* 2026 Jan;268(1):65-76. doi: 10.1002/path.6488. Epub 2025 Nov 6. PMID: 41199529.
- **Zhang J**, Ma B, Wang Z, Chen Y, Li C, Dong Y. Extracellular vesicle therapy for obesity-induced NAFLD: a comprehensive review of current evidence. *Cell Commun Signal.* 2024 Jan 9;22(1):18. doi: 10.1186/s12964-023-01292-0. PMID: 38195552; PMCID: PMC10775587.
- **Zhang J**, Bechard ME, Coffey RJ, Huh WJ. Lrig3 Suppresses EGFR Signaling to Prevent the Development of SPEN in a Manner Dependent on Lrig1. *CMGH.* (under revision)
- Moreno JE, **Zhang J**, Gabriel TT, Park J, Huh WJ. Targeting Notch and Hedgehog Signaling Reverses Ménétrier's Disease Associated Phenotypes. (Manuscript in preparation)
- **Zhang J**, Moreno JE, Ofosu FK, Huh WJ. Single-Cell RNA Sequencing Reveals Hepatocyte Dynamics Across Different Stages of High-Fat Diet Induced Disease Modeling. (Under Manuscript).

• **Presentations:**

- **Jiali Zhang**, Matthew E. Bechard, Robert J. Coffey, Won Jae Huh. Lrig3 Represses Spasmolytic Polypeptide-Expressing Metaplasia Formation in Chief Cells by Inhibiting EGFR Signaling. DDW, Chicago 2026.
- **Zhang, J.**, W.J.Huh. Lrig3 Represses Gastric Tumorigenesis by Inhibiting EGFR Signaling. 5th Annual Yale Postgraduate Symposium, 2025.
- **Zhang, J.**, Z. Wang, C. Li, and Y. Dong. Therapeutic Effect of Peony Seed Oil's Major Component (α -Linolenic Acid) on HepG2 Cell Models of Lipid Accumulation. *The 22nd Academic Symposium of the Animal Anatomy and Histology & Embryology Branch of the Chinese Association of Animal Science and Veterinary Medicine*, 2023.

LEADERSHIP EXPERIENCE

CAU Student Organization, *Member of Arts and Sports Department*, Beijing, China 2021-2022

- Led sponsorship acquisition and organized traditional Chinese festival performances for faculty and students

JLAU Student Organization, *Head of Arts Department*, Jilin, Chang Chun, China 2019-2020

- Led dance choreography and directed performance team for students' production
- Organized dance competitions for students, coordinated participants, logistics, and event execution
- Initiated and organized campus-wide singing competitions for students, drove event planning, promotion, and large-scale student engagement

JLAU Student Organization, *Member of Cultures and Sports Department*, Jilin, Chang Chun, China 2018-2019

- Led dance choreography for students
- Organized and managed a campus-wide sports meet for faculty and students

TEACHING AND INTERSHIP EXPERIENCE

Beijing Elk Ecological Experimental Animal Center, *Inter*, Beijing, China March 2023

- Collaborated on wildlife disease surveillance project examining CWD in cervid populations.

College of Veterinary Medicine, China Agricultural University, *TA*, Beijing, China 2021-2022

- Teaching assistant for undergraduate Animal Anatomy Laboratory. Prepared laboratory equipment assisted undergraduate students during anatomy lab sessions, and supported grading of assignments and exams.

Ruipai Pet Hospital, *Inter*, Chang Sha, Hu Nan, China April 2020

- Managed clinical cases including feline stomatitis treatment protocols and post-operative care.
- Mastered IV catheter placement on dogs and cats, and medication administration techniques.

SCHOLARSHIP AND AWARD (TOP 5%)

China Agricultural University, Beijing, China September 2021-June 2024

- Beijing Outstanding Graduates (2024)
- Outstanding Graduate of the University (2024)
- National Scholarship (2023) (**Top 1%**)
- Xiong Dashi Dean's Scholarship (2023)
- First-class Academic Scholarship (2023)
- Second-class Academic Scholarship (2021&2024)
- University Merit Student (2023)

Jilin Agricultural University, Chang Chun, Jilin, China September 2017-June 2021

- Second-Class major Scholarship (2021)
- Jilin Province Scholarship (2018&2020),
- DNB Encouragement Scholarship (2018)
- Excellence Award at the 6th Shengtai'er Cup Veterinary Medicine Professional Skills Competition (2019)

- First-class Major Scholarship (2017&2018&2019&2020)
- College Merit Student (2018)
- University Model Student in Learning (2018)

HOBBIES AND INTERESTS

- Professional Latin dancer (15 years)
- *Interests:* Dancing (cover all South American dance types), Resistance training (5 years), Piano, Bouldering, Top rope, Yoga, Meditation, Pilates, Swimming, Snorkeling, Hiking, Boxing, Skiing, Frisbee, Sauna, Board role-playing game, Badminton, Cooking, Sleeping, Watching movies, Tasting food, Helping the homeless on the street.